Implementation Security and Side-Channel Resistance in Post-Quantum Cryptography

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ABSTRACT

The first standards that are quantum-resistant cryptography primitives to protect digital infrastructure are the 2024 standards by the National Institute of Standards and Technology of FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) as part of the Sustainable Development Goals which are 9 and 11. This chapter discusses the essential discontinuity between the mathematical security assurance of these standards and their implementation security on embedded systems in smart city systems, industrial control networks and on systems in critical infrastructure. The Number Theoretic Transform of Z3329[X] over (X 2 56 + 1) which is the computational kernel of all three lattice-based standards forms a unique side-channel exposure profile using differential power analysis, cache timing, and fault-enabled chosen-ciphertext attacks. It is a chapter that analyzes the formal leakage models of countermeasure design, quantitatively maps the Number Theoretic Transform attack surface, arithmetic masking with performance benchmarks specific to each platform, and identifies four unsolved problems, the solution of which would be necessary to deploy in a sustainable way the post-quantum infrastructure that urban and industrial infrastructure demand.

Keywords: Number Theoretic Transform, ML-KEM, FIPS 203, Side-Channel Attacks, Differential Power Analysis, Arithmetic Masking, Barrett Reduction, Test Vector Leakage Assessment, Post-Quantum Cryptography, Smart City Security, Critical Infrastructure Protection, ISW Probing Model, Constant-Time Implementation, Embedded Systems Security, Fujisaki-Okamoto Transform

INTRODUCTION

Switching to quantum-resistant cryptography is not just a purely theoretical game in algorithm design new design - it is a pressing practical task of the digital infrastructure of sustainable development. Billions of endpoints in terms of networked devices include both sustainable development goal 9 (Industry, Innovation and Infrastructure) and SDG 11 (Sustainable Cities and Communities), which are IoT sensors in industries, smart grid controllers, autonomous vehicle communication systems, healthcare data networks, and municipal services platforms. All these endpoints run cryptographic protocols to verify the messages, come up with session keys and secure the data being transferred. The National Institute of Standards and Technology's 2024 publication of FIPS 203, FIPS 204, and FIPS 205 provides the algorithmic foundation for replacing the RSA and elliptic-curve cryptography that currently protects them, both of which are fully vulnerable to Shor's algorithm running on a cryptographically relevant quantum computer.

The algorithmic foundation is necessary but not sufficient. The security proofs underlying FIPS 203 (ML-KEM, the Module Lattice-Based Key Encapsulation Mechanism), FIPS 204 (ML-DSA, the Module Lattice-Based Digital Signature Algorithm), and FIPS 205 (SLH-DSA, the Stateless Hash-Based Digital Signature Algorithm) operate in the random oracle model with a black-box adversary that observes only inputs and outputs. A physical device is not a black box. A CMOS integrated circuit dissipates power

proportional to the Hamming weight of data transitions across its logic gates and buses. An adversary with a current probe on the power supply pin of an embedded controller observes a noisy but informative linear function of those transitions on every clock cycle. A microcontroller executing a branch instruction takes one code path or another, and the timing difference between those paths can be measured remotely through network timing oracles with sub-microsecond resolution. These observations are not modelled by the security reduction underlying ML-KEM's proof — they are explicitly excluded by its black-box adversary assumption — yet they provide direct information about the secret polynomial coefficients in ML-KEM's private key.

Three published results establish that the gap between proof-level security and implementation-level security is not hypothetical for post-quantum lattice schemes. Primas et al. (2017) demonstrated single-trace key recovery against a masked lattice-based implementation at CHES 2017, showing that even correctly masked schemes leak sufficient information for full key recovery from a single power measurement. Ravi et al. (2020) extended this to CRYSTALS-Kyber specifically using a convolutional neural network trained on traces from a separate device of the same model, recovering the full secret key from approximately 5,000 traces against a first-order masked implementation. Hermelink et al. (2021) demonstrated that compiler optimisation at the -O2 level can transform a correctly written constant-time decapsulation implementation into a non-constant-time implementation by converting conditional-move instructions into branches, opening a timing oracle that enables adaptive chosen-ciphertext key recovery with O(n log q) queries. In both instances, the attack went over a countermeasure that was properly put in place in the theoretical security model.

The magnitude of the deployment problem is measured in terms of the devices. The ARM Cortex-M4 processor - the core that pqm4 benchmarks are intended to be run on, as well as the embedded platform of choice to be analyzed in this chapter - drives an estimated 10 billion unit physical shipments in the industrial IoT, consumer electronics and infrastructure deployments. STMicroelectronics, the maker of the STM32 family based on Cortex-M4 core, is used in smart electricity meters, medical infusion pumps and automotive body control modules. Any of these devices executing post-quantum cryptographic protocols without providing an appropriate level of protection on the side channel serves as a possible key recovery target of a well-resourced adversary. The quantity of devices deployed in smart infrastructure applications today will continue to run when there could be cryptographically relevant quantum computers. The security of the implementation of the post-quantum cryptography on the latter devices is not something that could be postponed.

These concerns are magnified in a particular context of the smart city and critical infrastructure. Industrial IoT endpoints in SDG 9 contexts — sensor nodes on manufacturing floors, remote substations, pipeline monitoring equipment — are often physically accessible to adversaries and operate over lifetimes measured in decades. Smart city endpoints under SDG 11 — street-level air quality sensors, parking meters, public transit ticketing terminals, and utility meters — are installed in publicly accessible locations where power supply pins can be probed with inexpensive off-the-shelf equipment. Healthcare IoT devices protecting patient data under HIPAA and GDPR requirements face the additional constraint that the retroactive compromise enabled by inadequately protected post-quantum implementations could expose historical health records harvested before migration.

This chapter provides a structured account of the implementation security problem for post-quantum lattice-based cryptography, organised to serve both the research community and practitioners responsible for deploying quantum-resistant systems. The exposition proceeds from formal security model to physical attack to countermeasure to deployment platform. The Background section establishes the formal leakage models and reviews the existing literature on lattice-scheme side-channel attacks. The Main Focus of the Chapter section maps the Number Theoretic Transform attack surface quantitatively, derives the masking theory required for its defense, and identifies the composability failure at the arithmetic-Boolean interface that the standardisation process left unaddressed. The Solutions and Recommendations section analyses countermeasures based on platform specific performance information and realistic implementation advice to smart city and industry setting. Four open problems are listed in the Future Research Directions section,

all of which need to be solved before the field can boast of rigorous implementation security of post-quantum systems at the scale of smart city and SDG-infrastructure.

BACKGROUND

History The reinforcement of physical side-channel attacks That the encryption time of an RSA assembly is influenced by the secret key bit count led Kocher (1996) to remark that key recovers can be constructed using just time measurements. Kocher et al. (1999) later illustrated a study on the differentiability of power the device used in cryptographic computation, where the power used by a device in a crypto-computation is possibly correlated with theorized minuscule values and to extract that key by using statistical distinguishers in numerous exercises. These findings brought it to a formal conclusion that implementation security and algorithmic security are disjointed properties: a cryptographically sound algorithm may be implemented insecurely, and a cryptographically weak algorithm may be implemented by including physical side-channel security. The difficulty facing the post-quantum cryptography is that (algorithms based on) Lattices make new and undercharacterized implementation vulnerabilities, which are not adequately captured in the classical side-channel literature.

The t-Probing Model and Its Limits

The model that forms the basis of masking security is t-probing model of Ishai et al. (2003). In this model, a t -probing-secure circuit is one where the joint distribution of any t -probed values of the wires is statistically independent of the secret input. Their constructive result demonstrates that any Boolean circuit can be protected by encoding each secret value as t+1 shares, the secret x is split into $(x_0, x_1, \ldots, x_t)$ with $x_0 \oplus x_1 \oplus \ldots \oplus x_t = x$, where each share is independently uniformly random and the protected circuit has size $O(|C| \cdot d^2)$ where d = t+1 is the share count. Barthe et al. (2016) defined formal composability of the ISW gadgets in the EasyCrypt system, which allows mechanised verification of masking security of particular gadget compositions. These composability results are applied by automated verification tools such as maskVerif (Barthe et al., 2019) and SILVER (Knichel et al., 2020) which use circuit descriptions to perform a static analysis of these descriptions.

The insufficiency of the model of physical devices can be explained by the fact that the model is based on the assumption of the wire isolation. One of the probed wires in the model has one Boolean value; probing it will always give the value and no other. The transition between 0 and 1 of a voltage signal flowing in a wire in a real CMOS circuit dissipates charge Q = C -1 V 2 at the capacitance of the gate. The power trace at the power supply is a summation over all switches in the chip simultaneously of the Hamming distances in the switching activity of the chip in a single clock cycle, not just at a single wire but over the entire switching activity of the chip. The security guarantee of the t-probing model is thus not enough to define the security of masked implementation against the physical power measurement.

The Noisy Leakage Model

To fill the gap between the wire-level abstraction of the probing model and the overall physical reality of power measurement, Prouff and Rivain (2013) proposed the noisy leakage model. In this model every computation on a value x yields a leakage observation L = ℓ(x) + epsilon, where elli(·) is a deterministic leakage function - usually the Hamming weight HW(·) of CMOS devices - and epsilon is additive Gaussian noise of variance s 2. The main argument they make is that a first-order arithmetically masked calculation is safe on the noisy leakage model when 2 stret C · n, C is a constant which depends on the leakage-function and n is the number of independent operations. In an 896 Butterfly work is required to support a 256-coefficient Number Theoretic Transform with noise variance of the order of 896C. This condition is device-specific and cannot be verified from source code analysis alone.

The operational consequence is a mismatch between the theoretical guarantee and the practical evaluation. A first-order masked implementation satisfying the Prouff–Rivain noise condition is provably secure in the noisy leakage model. But the noise level σ^2 is a property of the measurement setup, not the implementation: the same implementation provides different effective security on a device whose power supply is filtered by a large bulk capacitor versus one with only a small surface-mount ceramic capacitor. A smart city sensor node on a regulated 3.3 V building management supply has a different noise profile than the same

microcontroller on an unregulated lithium cell in a street-level enclosure. Neither the noisy leakage model nor FIPS 203 provides guidance on determining whether a deployment environment's noise level satisfies the security condition.

Prior Work on Lattice Scheme Side-Channel Vulnerabilities

The first systematic side-channel analysis of lattice-based cryptography appeared with Bruinderink et al. (2016), who demonstrated a cache-timing attack against the BLISS lattice-based signature scheme using Flush+Reload on shared L3 cache. For Kyber specifically, Primas et al. (2017) demonstrated single-trace key recovery against a masked NewHope implementation, establishing that the Number Theoretic Transform's intermediate values leak sufficient information for full coefficient recovery from a single power measurement. Pessl et al. (2019) extended this with more practical single-trace attacks targeting the centered-reduction step in both NewHope and Kyber. Ravi et al. (2019) provided a systematic analysis of generic side-channel attacks applicable to all CCA-secure lattice-based KEMs, characterising the Fujisaki–Okamoto transform's re-encryption step as a particularly dangerous leakage source. Ueno et al. (2022) demonstrated the curse of re-encryption — that the re-encryption step required by the Fujisaki–Okamoto transform in all three NIST-standardised KEMs creates a structural leakage opportunity difficult to fully mask. Hermelink et al. (2021) showed that compiler optimisation transforms can undermine constant-time implementations at the source level.

The countermeasure literature has developed in parallel. Becker et al. (2022) provided the first formally verified first-order masked implementation of Kyber, establishing the composability argument required for the arithmetic-Boolean interface and demonstrating that the ISW multiplication gadget applied to $Z_{3}''_{2}‹$ arithmetic achieves 1-probing-security when composed with their specific B2A/A2B conversion gadgets. Coron et al. (2023) extended masked implementations to NTRU, providing a higher-order masking framework for a different post-quantum signature scheme that illustrates the general approach. Kannwischer et al. (2019) pqm4 benchmark suite gives the reference cost of the performance infrastructure on ARM Cortex-M4 hardware, and updates on optimisations to the pqm4 community, as FIPS 203 replaces older Kyber versions. Despite this progress, no second-order composable masking framework for ML-KEM exists in the published literature, and FIPS 203 specifies no side-channel test requirements for any assurance level.

The landscape of development tooling for post-quantum implementation security is expanding. The Open Quantum Safe project (OQS) maintains liboqs, an open-source library of post-quantum algorithm implementations including ML-KEM, designed for integration testing and performance benchmarking on desktop and server platforms. However, liboqs is explicitly not designed for hardware security: it makes no constant-time guarantees on all supported platforms, is not masked, and is documented as unsuitable for deployment in environments where physical side-channel attacks are a threat. The PQClean project provides cleaner, more portable implementations than the NIST reference code with some consideration for coding discipline, but similarly does not provide masking. The gap between research-grade masked implementations — such as those in Becker et al. (2022) and the pqm4 masked variants — and production-ready, independently verified, FIPS-conformant masked implementations suitable for smart city and industrial deployment remains large. Bridging this gap is as much an engineering and standardisation challenge as a research challenge.

MAIN FOCUS OF THE CHAPTER

Issues, Controversies, and Problems

The main issue that this chapter is focused on is the structural incompleteness of FIPS 203, FIPS 204, and FIPS 205 as deployment specifications. Individual standards are fully specified in algorithmic terms and are given formal security proofs, and each is silent about the physical security conditions needed to make the guarantee apply to deployments in smart city and industrial infrastructure. This silence indicates a real technical deficiency in the research literature: the composable security tools needed to specify physical security requirements of the complete ML-KEM computation have not yet been invented on the generality required. This problem is elaborated in finer technical detail in the following subsections.

The Number Theoretic Transform Attack Surface

ML-KEM, ML-DSA, and SLH-DSA all perform polynomial multiplication in the ring $R_v = Z_{3329}[X]/(X^{256}+1)$. The modulus q = 3329 was selected because $3329 - 1 = 3328 = 2^8 \times 13$, guaranteeing the existence of primitive 256th roots of unity modulo 3329 — a necessary condition for an efficient 256-point Number Theoretic Transform. The primitive 256th root of unity used in the reference implementation is $\zeta = 17$, satisfying $\zeta^{256} \equiv -1 \pmod{3329}$ and $\zeta^{512} \equiv 1 \pmod{3329}$. The forward Number Theoretic Transform proceeds in 7 layers. At each layer, 128 Cooley–Tukey butterfly operations execute in total. Each butterfly reads coefficients (a, b) and writes:

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a' = a + ζᵏ · b  (mod 3329)
b' = a − ζᵏ · b  (mod 3329)
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where ζ^k is drawn from a precomputed table of 128 twiddle factor values. Modular reduction in each butterfly uses Barrett reduction with the precomputed constant $m = \lceil 3^{26}/3329 \rceil = 20159$, replacing division by a multiply-and-shift: $\lfloor x/3329 \rfloor \approx (x \times 20159) >> 26$.

The side-channel exposure arises because the intermediate value $\zeta^k \cdot b \pmod{3329}$ is a deterministic linear function of the secret coefficient b. Since the twiddle factor ζ^k is public and fixed per butterfly position, $HW(\zeta^k \cdot b \bmod 3329)$ — the Hamming weight of the multiply result, proportional to power consumed by the SMULL instruction on ARM Cortex-M4 — is a known function of the unknown secret. The Pearson correlation distinguisher achieves probability-of-success p = 0.95 at measurement count $N \approx (Q^{-1}(0.05) \cdot \sigma / \rho)^2$. Empirical measurements on an STM32F407 using a Langer RF-U 5 electromagnetic probe give $\rho \approx 0.3$ to 0.4 for an unmasked NTT butterfly multiply, with $\sigma \approx 0.08$ in normalised units, yielding $N \approx 40$ to 70 traces per coefficient. Because the 256 secret coefficients can be attacked independently in parallel, full key recovery from a 256-coefficient secret polynomial requires approximately 60 decapsulation measurements — less than one second of physical access at a typical embedded decapsulation rate of 100 operations per second.

A second attack vector operates through the centered-reduction step, which maps NTT output coefficients from [0, 3328] to [−1664, 1664]. The reference C implementation performs this as a conditional subtraction: if (a > 1664) a -= 3329. This branch leaks the comparison outcome through both timing differences and cache eviction patterns. Pessl et al. (2019) demonstrated recovery of secret coefficients from a single decapsulation using the Flush+Reload cache side-channel technique of Yarom and Falkner (2014) applied to this branch on x86-64 processors with shared L3 cache. This branch has been removed by the ARM assembly implementations of pqm4 (Kannwischer et al., 2019) using an arithmetic right-shift idiom mask = (a >> 31) & 3329; a = a + mask that gives the correct centred result with no variation in taken/not-taken branches.

A third attack vector targets the Fujisaki–Okamoto transform's implicit-rejection step. ML-KEM decapsulation re-encrypts the recovered message and compares the resulting ciphertext c' byte-by-byte with the input ciphertext c. If this comparison terminates early upon finding the first byte mismatch rather than running to completion regardless of the match outcome, an adaptive adversary can distinguish matching from non-matching ciphertexts by timing the comparison. This timing oracle enables key recovery through a sequence of O(n log q) adaptive queries, as Hermelink et al. (2021) demonstrated. The threat is compiler transformation: Cauligi et al. (2020) demonstrated at ACM PLDI 2020 that GCC 9.3 and LLVM/Clang 10.0 at optimisation level -O2 both transform constant-time C source patterns into branching code sequences under cost-model conditions that apply to Cortex-M4 targets.

Masking Theory Applied to Polynomial Arithmetic

First-order arithmetic masking of a value $x \in Z_{3329}$ decomposes it as a pair (x_0, x_1) where x_0 is chosen uniformly at random from Z_{3329} and $x_1 = x - x_0 \bmod 3329$. The joint distribution of the pair is uniform over all solutions to $x_0 + x_1 \equiv x \pmod{3329}$; probing either share individually yields a uniformly random value independent of x. Applied to the 256-coefficient secret polynomial $s \in R_v$, first-order masking requires 512 scalar values per polynomial. For ML-KEM-768 with three secret polynomials, a complete first-order masking requires $6 \times 256 = 1{,}536$ independent uniform random values from Z_{3329} per decapsulation — 3,072 bytes of fresh masking randomness generated using SHAKE256 on devices without a hardware RNG.

Point-wise polynomial multiplication in the NTT domain requires computing $H_i = F_i \cdot G_i \pmod{3329}$ for each of 256 coefficient positions, where both F_i and G_i are masked. The ISW multiplication gadget for first-order arithmetic masking (Ishai et al., 2003; adapted to prime fields by Coron et al., 2012) uses four scalar multiplications and one fresh random field element r per coefficient pair, computing: $H_{i,0} = F_{i,0} \cdot G_{i,0} + F_{i,0} \cdot G_{i,1} + r \pmod{3329}$ and $H_{i,1} = F_{i,1} \cdot G_{i,0} + F_{i,1} \cdot G_{i,1} - r \pmod{3329}$. The gadget is 1-probing-secure: any single probed intermediate is statistically independent of H_i. At 256 coefficient positions both masked and unmasked computational costs are 1,024 scalar multiplications and 256 new random numbers respectively, and without considering the overhead of converting between Boolean and Arithmetic representations a 4-fold multiplication count increase is realized.

The Composability Failure at the Arithmetic-Boolean Interface

The greatest structural issue in the implementation of masked ML-KEM is that the two computational domains that the scheme uses are incompatible. The centered binomial distribution sampling step - that computes secret and error polynomials coefficients as the difference between popcount operations on output bytes of SHAKE256 - is naturally expressed, in terms of bitwise Boolean operations on GF(2). The Number Theoretic Transform subsequently processes these coefficients using modular arithmetic over $Z_{332<}$. A masked implementation must cross this boundary using Boolean-to-Arithmetic (B2A) conversion.

Goubin's B2A algorithm (2001) converts a first-order Boolean-masked value (x_0, x_1) with $x_0 \oplus x_1 = x$ to an arithmetic-masked value (x_0', x_1') with $x_0' + x_1' \equiv x \pmod{2^k}$ using one fresh random value and five arithmetic operations. For k = 12 (since $2^{12} = 4096 > 3329$), one reduction step modulo 3329 is additionally required. Applied to all 256 coefficients of three secret polynomials in ML-KEM-768, the B2A conversion contributes approximately $256 \times 3 \times 6 = 4{,}608$ scalar arithmetic operations, representing 12 to 18% of the total masked overhead above the unmasked baseline.

The greater issue is composability. The composability theorem of the ISW states that all of the subcircuits at a composition boundary are based on the same algebraic ring. This is not satisfied when arithmetic-masked values above Z 33 2 L are operated with Boolean-masked operations on GF(2). Becker et al. (2022) stated this issue and offered a formal demonstration that their B2A/A2B gadget composition maintains first-order probing security - although only in their gadget selections and may need re-testing with other implementations. There is no overall framework of second-order or higher masked ML-KEM that is provided in the published literature. This gap poses a direct constraint to the security assurances that can be used in smart city and industrial implementations that need Common Criteria EAL5+ or equivalent assurances.

SOLUTIONS AND RECOMMENDATIONS

Masking Performance on Embedded Targets

Table 1 shows counts of cycles when using all three ML-KEM parameter sets when unmasked, first-order and second-order masked are used on an STM32F407VGT6 at 168 MHz with instruction caches enabled. The unmasked benchmarks come pqm4 (Kannwischer et al., 2019), first-order masked benchmarks can be found in Becker et al. (2022). The count of second-order cycles is projected based on the scaling of ISW gadget complexity (d 2 product of share-product cost) at masking order d and the three-share B2A conversion and expressed as ranges since there were no published second-order ML-KEM results at the time of writing.

Table 1. ML-KEM cycle counts on ARM Cortex-M4 at 168 MHz (thousands of cycles)

Variant	Operation	Unmasked	1st-Order	2nd-Order (est.)	1st Overhead	2nd Overhead (est.)
ML-KEM-512	KeyGen	452	1,620	5,100–5,700	3.6×	11.3–12.6×

	Encaps	506	1,870	5,800–6,500	3.7×	11.5–12.8×
	Decaps	533	2,010	6,200–7,100	3.8×	11.6–13.3×
ML-KEM-768	KeyGen	648	2,380	7,300–8,200	3.7×	11.3–12.7×
	Encaps	697	2,620	8,100–9,100	3.8×	11.6–13.1×
	Decaps	737	2,850	8,900–10,100	3.9×	12.1–13.7×
ML-KEM-1024	KeyGen	888	3,280	10,100–11,400	3.7×	11.4–12.8×
	Encaps	951	3,520	10,900–12,300	3.7×	11.5–12.9×
	Decaps	1,013	3,990	12,400–14,100	3.9×	12.2–13.9×

Source: Unmasked from Kannwischer et al. (2019); 1st-order masked from Becker et al. (2022); 2nd-order are author projections from ISW gadget complexity scaling.

The 3.7 to 3.9× overhead at first order reflects the ISW gadget's 4× multiplication cost and B2A conversion overhead. The random number generation cost — drawing 3,072 bytes of uniform random data per ML-KEM-768 decapsulation using SHAKE256 on a device without a hardware RNG — contributes approximately 10 to 15% of the total first-order overhead on STM32F407 hardware. Devices equipped with a hardware true random number generator, such as the STM32's built-in RNG peripheral, reduce this contribution to negligible levels by parallelising random generation with computation. The masking overhead scales differently with parameter set than the unmasked computation: masking cost is dominated by the ISW multiplication gadget applied to point-wise NTT-domain products, which grows as the square of the polynomial rank k scaled by 256 coefficients, while the unmasked NTT cost grows linearly in k. For a FIDO2 hardware token with a 300 ms response budget, first-order masked ML-KEM-768 decapsulation at 2.85 million cycles on a 168 MHz processor completes in approximately 17 ms. Second-order masking at approximately 10 million cycles completes in approximately 60 ms. Both are deployable; the choice depends on the physical access model and certification requirements.

Shuffling

Shuffling randomises the execution order of independent operations within each Number Theoretic Transform layer to prevent trace alignment. The 128 butterfly operations within a single NTT layer are data-independent and can be executed in any permutation without affecting the mathematical output. A Fisher–Yates shuffle applied to a 128-element permutation array before each layer execution, using a pseudorandom seed from a hardware RNG or SHAKE256, costs O(128) random bytes per layer and O(1) additional arithmetic per butterfly. Veyrat-Charvillon et al. (2012) established that shuffling n independent operations reduces single-trace signal-to-noise ratio by a factor proportional to $\sqrt{n}$, providing approximately $(1/2) \log_2 n$ bits of effective noise. For n = 128 (one NTT layer), this yields approximately 3.5 bits of effective noise reduction — meaningful but not sufficient in isolation, as it does not prevent multi-trace correlation attacks. Shuffling works best in combination with first-order masking, in which it increases the effective trace count needed to perform a second-order profiling attack by compelling adversarial alignment within the shuffled NTT layer.

Constant-Time Implementation Discipline

Timing observation is a precondition which requires security in any threat model that involves timing observation. An implementation of masking using a secret-dependent timing leak offers the protections of masking to power channel attacks, and is entirely vulnerable to a remote timing attacker. Practically, this problem is that C source code optimisation compilers do not maintain constant-time discipline. Relational verification of constant-time properties is done at the LLVM level of the intermediate representation with

ct-verif tool (Almeida et al., 2016) that verifies that no control flow or memory access decision is reliant on a specified secret input. The runtime statistical testing is offered by the dudect tool (Reparaz et al., 2017): the TVLA t-test is used to the timing of the execution as opposed to power traces and anonymous timing changes in secrets are identified without physical measurement instrumentation. The pqm4 ARM implementation of ML-KEM is able to pass the dudect test. The FIPS 203 does not require any of the tools. Independent of the renderer on which ML-KEM is executed on a platform other than ARM, practitioners executing it on that platform are encouraged to ensure constant-time aspects at both compile-time and runtime using ct-verif and dudect, respectively, after which it should be deployed.

Platform-Specific Countermeasure Analysis

Each hardware platform will have optimal countermeasure strategy, which cannot be addressed by a single implementation. Table 2 summarises the attack surface and protection recommended to 5 representative deployment context related to the realistic smart city landscape and critical infrastructure landscape.

Table 2. Recommended countermeasure and attack surface based on side channel attack by deployment platform.

Platform	Primary Attack	NTT Leakage	Recommended Protection	SDG Application Context
ARM Cortex-M4 (STM32F407 class)	Power via supply pin; near-field EM probe	HW of SMULL butterfly register writes	1st-order masking + shuffling; 2nd-order for EAL5+	Smart city IoT sensors; smart meters; industrial control endpoints
ARM Cortex-A (application processor)	Shared L2/L3 cache timing; power	Branch-dependent cache eviction in centered reduction	Constant-time assembly; 1st-order masking; process isolation	Smart city gateways; urban mobility platforms; e-health servers
FPGA (Artix-7 class)	Aggregate HW of parallel butterflies; glitch leakage	Combinational glitch in butterfly logic before output register	Threshold implementation or domain-oriented masking at gate level	Industrial SCADA; smart grid protection relays; infrastructure HSMs
HSM FIPS 140-3 Level 3	Fault injection; EM through active shielding	Fault in FO comparison; twiddle factor loading fault	Hardware masking + tamper detection + fault detection codes	National PKI; critical infrastructure CA; financial settlement
Smartcard / Secure Element	Close-range power/EM contact	HW of NTT multiply; FO comparison timing	2nd-order minimum per CC EAL5+; hardware RNG required	Digital identity; e-healthcare records; smart city access control

Source: Attack characterisation from Becker et al. (2022), Karabulut and Aysu (2021), Hermelink et al. (2021), and Ravi et al. (2020).

Field-programmable gate array implementations present a fundamentally different leakage structure from software. A pipelined 256-point NTT with 128 butterfly units operating in parallel produces power traces in which each clock cycle reflects the aggregate Hamming weight of 128 intermediate values. Karabulut and Aysu (2021) characterised this for Kyber on a Xilinx Artix-7 and found that combinational glitch logic in unprotected butterfly circuits produces first-order TVLA-detectable leakage before the multiplier output is registered. Addressing glitch leakage requires threshold implementations (Nikova et al., 2006) or domain-oriented masking (Gross et al., 2016) at the gate level, with 4 to 6× gate count overhead for first-order protection.

Test Vector Leakage Assessment

Test Vector Leakage Assessment (TVLA), introduced by Goodwill et al. (2011), uses Welch's t-test to compare power trace distributions between fixed-input and random-input measurements. The null hypothesis is equal means; a large $|t|$ statistic indicates that power consumption carries information about the input. The standard threshold $|t| > 4.5$ corresponds approximately to a false-positive probability of 3.4×10^{-6} per sample point under Gaussian trace distributions. The pqm4 ARM assembly implementation of ML-KEM-768 passes 100,000-trace first-order TVLA on STM32F407 hardware.

The existence of first-order linear leakage at the tested number of traces at TVLA passing certifies the absence. It was formally determined by Standaert et al. (2009) that TVLA cannot certify security: a device that passes an initial test with 100,000 traces on the first order may still be vulnerable to a higher-order attack with 500,000 traces on an improved measurement apparatus, or may be vulnerable to a second-order distinguisher on covariance between pairs of sample points. In the case of smart cities and industrial implementations, the testing of TVLA can be considered a necessity, but not a sufficient state: it eliminates the possibility of first-order leakage, but does not prove the elimination of higher-order vulnerabilities that an adversary with physical access would use.

Practical Deployment Guidance for Smart City and Industrial Applications

To translate the analysis of the countermeasures into an actionable advice related to the smart cities projects, one will have to follow a tiered approach based on the protection levels and the access risk associated with the physical aspect. Where the endpoint of the IoT is physically secure, such as sensor board automation of the indoor building, and the physical boundaries conveniently ensure the safety of the physical equipment, first-order arithmetic masking and constant-time ARM assembly are a sufficient starting point. The resulting 3.7 to 3.9x performance can be supported by sensor node duty cycles and dudect timing tests ought to be implemented on every build of the firmware as part of the continuous integration process. A second-order masking must be mandated on the endpoints of publicly accessible sites - street sensors, transit terminals, utility meters available outside the building, and enclosures must have a seal that is tamper-evident. A measurement study of the deployment location's electromagnetic environment is warranted before finalising the countermeasure specification. For critical infrastructure endpoints — grid protection relays, SCADA communication gateways, supervisory control nodes — FIPS 140-3 Level 3 certified HSMs provide the appropriate baseline, with independent evaluation of HSM firmware for the Fujisaki–Okamoto fault attack pathway identified by Hermelink et al. (2021).

FUTURE RESEARCH DIRECTIONS

Four open problems require resolution before the field can claim rigorous implementation security for post-quantum lattice-based cryptography at the scale that smart city and critical infrastructure deployment demands.

Composable Masking Frameworks for Mixed-Domain Polynomial Arithmetic

The first-order probing security of masked ML-KEM decapsulation is established by Becker et al. (2022) for their specific gadget composition. No general composable framework exists for masked NTT computations beyond first order. The mathematical obstacle is the arithmetic-Boolean interface: the NTT operates over $Z_{332‹}$, sampling and serialisation operate over Z_{28}, and the Fujisaki–Okamoto comparison is a Boolean function on byte arrays. A fully composable second-order security proof would require a composition theorem handling transitions between arithmetic and Boolean masking domains at the required generality. Developing such a framework requires combining lattice-scheme analysis tools with the formal verification infrastructure of EasyCrypt and maskVerif — a combination that has not yet been achieved. This gap directly limits security guarantees available for EAL5+ certified smart city infrastructure.

Compiler-Verified Constant-Time Guarantees for Post-Quantum Implementations

Cauligi et al. (2020) documented 17 code patterns in production cryptographic libraries where constant-time source-level implementations became non-constant-time after compilation at -O2. Future work should establish: mandatory constant-time verification as a condition of NIST Cryptographic Module Validation

Program validation for implementations at any assurance level; a standardised test suite using dudect with specified statistical power for ML-KEM timing tests; and a formally verified C implementation — analogous to HACL* for Curve25519 and ChaCha20-Poly1305 — whose constant-time properties are proved to survive compilation on the three dominant embedded compiler toolchains (GCC, LLVM, and IAR EWARM). The HACL* approach using the F* proof assistant and the Vale/Jasmin assembly verification frameworks provides the methodological template.

NIST Side-Channel Test Guidance for FIPS 203, 204, and 205

FIPS 203, FIPS 204, and FIPS 205 specify Known-Answer Tests for correctness and nothing for physical security. Most commercial and industrial implementations will pursue FIPS 140-3 Level 1 or Level 2 validation, which is entirely silent on side-channel properties. ISO/IEC 17825:2016 specifies TVLA for cryptographic module testing but predates ML-KEM and contains no NTT-specific guidance. The NIST National Cybersecurity Center of Excellence and ISO/IEC JTC 1/SC 27 should jointly develop: NTT-specific TVLA test inputs specifying which butterfly positions and polynomial dimensions constitute primary leakage targets; required masking orders for specified assurance levels correlated with smart city and industrial deployment scenarios; and minimum trace counts for TVLA tests at each assurance level. This standards work would provide the clear, implementer-accessible specification that the current gap in FIPS 203 creates.

Tight Information-Theoretic Leakage Bounds for ML-KEM's NTT

No tight bound exists on the mutual information $I(s_i; HW(\zeta^k \cdot s_i \bmod 3329))$ between a secret coefficient s_i drawn from ML-KEM's centered binomial distribution η_2 and the observable power leakage from the butterfly multiply. A tight such a bound would enable tight statements of the form: the benefit of any DPA adversary in the case of first-order masked ML-KEM-768 is at least $\varepsilon(\sigma\ 2, N, T)$ where N is the number of traces and T is the adversary computation time. To compute this bound, one has to evaluate the mutual information of all the 256 butterfly positions, and all three ML-KEM parameter sets, which is a combination of lattice-scheme analysis and the leakage quantification tools of the side-channel community. This was a connection that would be missing between the theoretical model of noisy leakage and the empirical results of the attacks in the published literature, making it possible to select the countermeasures principled in both the implementation of smart cities and industries without necessarily referring to the empirical outcomes of the specific hardware.

The migration post-quantum calendar adds further panic to the implementation security research. The NIST FIPS 203 standard was released in August 2024, and the national security systems are mandated by the NSA guidance, Commercial National Security Algorithm Suite 2.0 (CNSA 2.0.), to be fully migrated to ML-KEM by 2030. In the case of SDG 9 infrastructure and SDG 11 infrastructure, this is an aggressive timeline: industrial control systems in smart city implementations have firmware update cycles of years not months, and the embedded security modules supply chain, smartcards, secure elements, and hardware security modules involved has lead times and certification times of two to three years between the finalisation of cryptographic algorithms and the delivery of certified hardware. Organisations responsible for critical infrastructure cannot wait until a cryptographically relevant quantum computer has been demonstrated to begin the implementation security evaluation process; by that point, adversaries will already have harvested encrypted traffic for retroactive decryption. The harvest-now-decrypt-later threat model means that inadequately protected ML-KEM implementations deployed today represent a risk that materialises retroactively, after the fact. Implementation security evaluation must therefore be treated as part of the initial deployment decision, not as a post-deployment audit.

The interaction between post-quantum implementation security and post-quantum key distribution protocols presents a further underexplored research direction. Hybrid classical-post-quantum key establishment protocols — which are recommended as a transitional measure by NIST, NSA, ETSI, and BSI — combine classical ECDH with ML-KEM in a single handshake, providing security if either component is secure. From an implementation security perspective, hybrid protocols create new side-channel considerations: the combined handshake involves both elliptic-curve scalar multiplication and NTT polynomial multiplication in the same execution context, and the power trace of a hybrid handshake

potentially leaks information about both components simultaneously. Whether this combined leakage is strictly more dangerous than either component alone depends on how the two computations are scheduled and whether they are masked independently or with a shared masking strategy. No published analysis of hybrid protocol side-channel leakage exists at the time of writing.

Beyond these four specific open problems, the SDG-relevant deployment contexts raise additional research priorities. The energy efficiency of post-quantum cryptographic operations on battery-powered smart city sensor nodes — where the masking overhead of 3.7× in cycle count translates directly to 3.7× in energy consumption — creates a fundamental tension between security and sustainable operation that existing countermeasure designs have not addressed. Lightweight masked implementations of ML-KEM on Class 1 IoT devices with limited memory and/or limited computation are a research topic. Formal verification of hardware security module firmware of post-quantum algorithms, e.g. Arm TrustZone with ProVerif or Tamarin security protocol models, is not well-explored. Lastly, the case of implementing quantum key distribution and post-quantum cryptography hybrids to use in backbone infrastructure of a smart city has put new side-channel considerations on hybrid protocol handshakes that are not examined.

CONCLUSION

This chapter has followed the disjunction between the security assurance of ML-KEM and the physical security of the implementation of the latter, specifically within the scope of the smart city and of critical infrastructure where the ramifications of poor implementation security are most severe. The structural gap lies in the fact that MLWE security proofs are run in a model that does not involve physical observations, and the Number Theoretic Transform butterfly computation that is useful in making ML-KEM efficient can be physically measured - the values generated by the intermediate can be linear functions of the coefficients of the secret polynomials, and it is directly measurable by the power consumption of CMOS devices.

The practical severity of the gap is determined by three outcomes of the published literature. Primas et al. (2017) obtained the single-trace key recovery against a masked lattice scheme. Ravi et al. (2020) applied this specifically to CRYSTALS-Kyber, and they used a deep-learning profiling attack that circumvents first-order masking using about 5,000 traces. Hermelink et al. (2021) demonstrated that, even with correctly written decapsulation implementation, optimisation of the compiler can unintentionally turn on a timing channel allowing adaptive chosen-ciphertext key recovery without any power measurement equipment. FIPS 203 has no solutions to these vulnerabilities.

The implementation security has to be considered against the threat model that is defined by the deployment situation. A smart city parking meter mounted in an open area is not adversarial of the same smart meter mounted in a fenced high-voltage compound in a sub-station protection relay. Both can execute ML-KEM to provide authenticated session setups, and the adversary of meter may execute a period of sustained measurement with electromagnetic sensors during weeks or months, whereas the adversary of relay has only physical security boundaries that restrict access time and equipment. This should be modeled in the countermeasure specification: first-order masking with shuffling is suitable to the relay, where the adversary is able to access the device very infrequently and very briefly (physical access); and second-order masking is suitable to the parking meter, where the adversary can access the device as often as they can and as often as possible (physical access). This threat-model-to-countermeasure mapping is not offered in FIPS 203 and there is currently no NIST guidance document that provides SDG-infrastructure deployers with the tiered specification that they require.

The countermeasure landscape offers practical protection. First-order arithmetic masking at 3.7 to 3.9× cycle count overhead is deployable on ARM Cortex-M4 class devices within the performance budgets of smart city IoT applications. Shuffling provides additional single-trace protection at negligible cost. Constant-time assembly implementations verified with dudect eliminate the timing attack surface. Hardware security modules at FIPS 140-3 Level 3 and above address residual risks through physical tamper detection. What the landscape does not provide — a composable second-order masking framework, compiler-verified constant-time guarantees, standardised NIST side-channel test requirements, and tight leakage bounds — represents the research agenda that the post-quantum cryptography and side-channel

communities must address jointly. The SDG 9 and SDG 11 infrastructure that post-quantum cryptography is meant to protect — and the populations whose digital rights, economic participation, and physical safety depend on that infrastructure — cannot afford to wait for those gaps to close on their own.

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KEY TERMS AND DEFINITIONS

Arithmetic Masking: A secret-sharing countermeasure against power analysis attacks in which a secret value x over a prime field Z_v is split into d shares ($x_0, x_1, \ldots, x^d_{-1}$) such that $x_0 + x_1 + \ldots + x^d_{-1} \equiv x \pmod q$ and each individual share is uniformly random and independent of the secret.

Barrett Reduction: An algorithm for computing modular reduction x mod q using precomputed constants that replaces integer division by a multiply-and-shift operation. For ML-KEM with q = 3329, the constant m = 20159 enables computation of $\lfloor x/3329 \rfloor$ as $(x \times 20159) >> 26$.

Differential Power Analysis: A statistical side-channel attack technique introduced by Kocher et al. (1999) that correlates the power consumption of a cryptographic device with hypothesised intermediate values to recover secret key material across many measurements.

Fujisaki-Okamoto Transform: A generic construction that converts a chosen-plaintext-secure public-key encryption scheme into a chosen-ciphertext-secure key encapsulation mechanism by re-encrypting the decapsulated plaintext and comparing the result with the input ciphertext. The byte-level comparison must execute in constant time to prevent timing-based key recovery.

ISW Probing Model: The formal security model of Ishai, Sahai, and Wagner (2003) in which circuit security is defined by the statistical independence of any t simultaneously probed wire values from the secret input. A circuit is t-probing-secure if no combination of t probed wire values provides information about the secret.

Module Learning With Errors: The lattice-based computational problem underlying ML-KEM's security. An adversary must distinguish between uniformly random matrix-vector pairs and pairs of the form (A, As + e) where s is a secret vector and e is a small-coefficient error vector. No known classical or quantum algorithm achieves sub-exponential advantage for this problem.

Number Theoretic Transform: A discrete Fourier transform computed over a finite field Z_v rather than the complex numbers. For ML-KEM, the 256-point NTT over $Z_{332‹}$ with primitive root $\zeta = 17$ converts polynomial multiplication in $Z_{3}''_{2‹}[X]/(X^{256}+1)$ into coordinate-wise multiplication in the transform domain, reducing $O(n^2)$ to $O(n \log n)$ operations.

Side-Channel Attack: An attack that exploits information leaked through the physical implementation of a cryptographic algorithm — including power consumption, electromagnetic emissions, execution timing, and acoustic emanations — rather than weaknesses in the mathematical algorithm itself.

Test Vector Leakage Assessment: A statistical methodology introduced by Goodwill et al. (2011) that uses Welch's t-test to compare the distribution of power traces between fixed-input and random-input measurements. A test statistic $|t| > 4.5$ at any sample point indicates measurable leakage at that point in the trace.

t-Probing Security: A circuit security property in which the joint distribution of any t simultaneously probed wire values is statistically independent of the secret circuit input. The ISW compiler constructs circuits satisfying this property by encoding each wire value as t+1 additive shares, with each cross-product intermediate protected by a fresh random value.